

Myke Vital Sao Temgoua
Department of Physics, Faculty of Science
University of Yaoundé I
P.O. Box 812, Yaoundé, Cameroon
myke-vital.sao@facsciences-uy1.cm

September 11, 2026

Prof. Kenneth M. Merz, Jr., Editor-in-Chief
Journal of Chemical Information and Modeling
American Chemical Society

Dear Prof. Merz,

We submit our manuscript “**Estimand Divergence Between Static Docking and Short Molecular Dynamics as a Triage Filter for African Antimalarial Natural Products**” for consideration as an Article in the *Journal of Chemical Information and Modeling* (JCIM).

Using a curated collection derived from 396 African natural products and 454 synthetic antimalarials, we evaluated a multi-target docking framework across 136 wild-type and mutant receptor states of PfDHFR and PfCRT. On a primary candidate cohort ($n = 12$), a novel Resistance-Resilience Score (RRS) provided a five-tier stratification decoupling predicted wild-type binding strength from mutant-state score retention. Explicit-solvent molecular dynamics (MD) simulations revealed a directional divergence in 87.5 % (7/8) of matched candidate–target states, wherein dynamic pocket relaxation compensated for static grid-docking penalties. This structural stress test demonstrates that static docking scores and short MD trajectories provide complementary, non-equivalent triage metrics for natural product lead optimization.

We present three major methodological contributions: (i) conceptualizing *Estimand Divergence* between static docking RRS and explicit-solvent MD pose retention as a positive triage filter; (ii) establishing a denominator-unbiased RRS framework that excludes non-binding wild-type baselines ($|S_{WT}| < 5.0 \text{ kcal mol}^{-1}$); and (iii) validating protocol robustness across an independent external sensitivity panel of 39 compounds (312 docking complexes) confirmed by GNINA convolutional neural network (CNN) affinity scoring with 100 % Class-A agreement. Grounded in the JCIM molecular dynamics reporting guidelines established by Soares et al. (2023), short explicit-solvent trajectories (10 ns) function effectively as secondary structural stress tests.

Code, source data, analysis scripts, and model inputs are available on GitHub under an open-source licence; a permanent Zenodo snapshot will be archived upon publication. Statistics are reported with effect sizes, permutation p -values, bootstrap confidence intervals, and Bonferroni-Holm adjustments.

Suggested reviewers.

1. Dr. Fidele Ntie-Kang, University of Buea — African natural-product computational discovery
2. Dr. Kelly Chibale, University of Cape Town — antimalarial drug discovery
3. Dr. John D. Chodera, Memorial Sloan Kettering — free-energy methods and MD best practices

This manuscript is original, not under consideration elsewhere, and all co-authors have approved the submission.

Sincerely,

Myke Vital Sao Temgoua
(on behalf of all co-authors)